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Swati M

B.Tech CSE Student

Education

SRM Institute of Science and Technology *Aug 2023 – May 2027*
B.Tech in Computer Science and Engineering *CGPA: 9.72/10.00*
Kattankulathur, Chennai

Skills

Languages	Python, C, SQL
Core CS	Data Structures and Algorithms, Computer Networks, OOPS, Operating Systems, DBMS
Cloud & Databases	AWS, MySQL
Tools	Git, GitHub, VS Code, Jupyter Notebook
Web Technologies	HTML, CSS
Libraries & Frameworks	NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, Transformers

Experience

Samsung PRISM Research Project *Mar 2025 – Sept 2025*
Audio-Visual Deepfake Detection Leveraging Digital Biometrics

- Developed a multimodal deepfake detection framework using audio-visual analysis and biometric features.
- Integrated Wav2Vec2, Vision Transformers, and cross-modal attention mechanisms for fake media detection.
- Worked on physiological signal analysis, audiovisual synchronization, and explainable AI to improve detection accuracy.

Infosys Springboard Internship 6.0 *Nov 2025 – Jan 2026*
ML-Driven Web Platform for Cryptocurrency Price Forecasting *Virtual*

- Built a machine learning-based web platform to analyze and forecast cryptocurrency price trends.
- Worked on data preprocessing, predictive modeling, and visualization of market insights.
- Implemented forecasting techniques to improve prediction accuracy and user interaction.

Projects

Stacked Multimodal Analysis for Parkinson's Detection
- Developed a multimodal framework combining speech and handwriting biomarkers for Parkinson's disease detection.
- Implemented a stacked SVM-based approach combining spectral, temporal, and handwriting features for improved classification performance.
- Applied preprocessing, feature extraction, and multimodal fusion techniques to enhance early-stage disease prediction accuracy

Crowdfunding Platform for Student Innovations
- Designed the backend database and workflows for a student project crowdfunding platform using MySQL.
- Implemented CRUD operations, joins, stored procedures, triggers, and indexing for efficient data handling.
- Structured the schema to support project listings, contributor tracking, and funding milestone management.

Brain Tumor Detection Using CNN with GradCAM Explainability
- Built a CNN-based model to classify brain MRI images into Tumor and No Tumor categories using TensorFlow and OpenCV.
- Integrated GradCAM explainability to generate visual heatmaps highlighting tumor regions for interpretability.

StoryMorph – AI-Powered Story Generator from Children's Drawings
- Built a Streamlit web app that converts children's drawings into short moral-based stories.
- Used BLIP image captioning to interpret drawings and GROQ LLM to generate coherent, child-friendly narratives.
- Designed an intuitive kid-friendly interface focused on simplicity, engagement, and age-appropriate storytelling.

Certifications

AWS Certified Cloud Practitioner	March 2026
AWS Certified AI Practitioner	March 2026
NPTEL – Introduction to Machine Learning	September 2025
NPTEL – Programming in Java	October 2024
Salesforce Certified Agentforce Specialist	December 2025